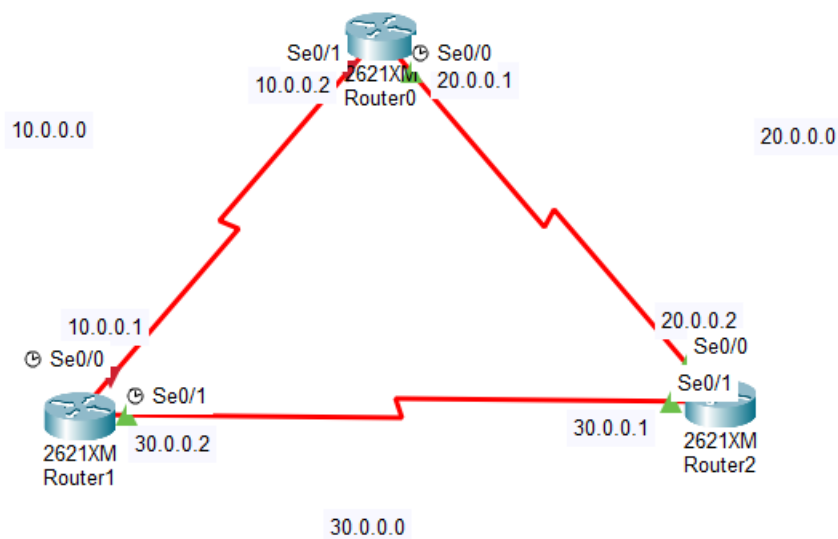


Day 4: Introduction to OSPF Theory: How OSPF works: Areas, LSAs, and cost.
Practical: Observe OSPF adjacency formation in Packet Tracer. Task: Write down steps to configure OSPF in a network.

Task 1.

Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.



Task 2.

Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

Router 1:

```
Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   administratively down down
FastEthernet0/1 unassigned      YES unset   administratively down down
Serial0/0       20.0.0.1        YES manual up           up
Serial0/1       10.0.0.1        YES manual up           up
Router#
Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 22/25/32 ms
Router#
```

Router 2:

```
Router#  
Router#  
Router#ping 20.0.0.2  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 33/37/47 ms  
  
Router#ping 30.0.0.1  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 30.0.0.1, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/19/23 ms
```

Router 3:

```
Router#ping 30.0.0.2  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/14/16 ms  
  
Router#ping 10.0.0.1
```

From	Command	Expected result
Router0	ping 10.0.0.1	Successful
Router0	ping 20.0.0.2	Successful
Router1	ping 30.0.0.1	Successful
Router1	ping 20.0.0.1	Successful
Router2	ping 10.0.0.2	Successful
Router2	ping 30.0.0.2	Successful

Task 3.

Configure OSPF on each router using the `router ospf 1` command and the `network` command to advertise all directly connected networks.

```

Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#10.0.0.0 0.0.0.255 area 0
^
% Invalid input detected at '^' marker.

Router(config-router)#10.0.0.0 0.255.255.255 area 0
^
% Invalid input detected at '^' marker.

Router(config-router)#net 10.0.0.0 0.0.0.255 area 0
Router(config-router)#net 20.0.0.0 0.0.0.255 area 0
Router(config-router)#net 30.0.0.0 0.0.0.255 area 0
Router(config-router)#
Router(config-router)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
00:33:05: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.2 on Serial0/0 from LOADING to FULL, Loading Done

00:34:26: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.3 on Serial0/1 from LOADING to FULL, Loading Done

```

Router 2:

```

router ospf 1
router-id 1.1.1.2
log-adjacency-changes
network 10.0.0.0 0.0.0.255 area 0
network 20.0.0.0 0.0.0.255 area 0
network 30.0.0.0 0.0.0.255 area 0
!
ip classless

```

Router 3:

```

!
router ospf 1
router-id 1.1.1.3
log-adjacency-changes
network 10.0.0.0 0.0.0.255 area 0
network 20.0.0.0 0.0.0.255 area 0
network 30.0.0.0 0.0.0.255 area 0
!
ip classless
!

```

Task 4.

Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router.

Hint: Look for the FULL/DR or FULL/BDR state to confirm adjacency.

Router 1

```
Router#  
Router#show ip ospf ne  
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.2	0	FULL/ -	00:00:31	20.0.0.2	Serial0/0
1.1.1.3	0	FULL/ -	00:00:36	10.0.0.2	Serial0/1

```
Router#
```

Router 2

```
Router#show ip ospf ne  
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:37	20.0.0.1	Serial0/0
1.1.1.3	0	FULL/ -	00:00:34	30.0.0.2	Serial0/1

```
Router#
```

Router 3

```
Router#show ip ne  
Router#show ip nei  
Router#show ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:30	10.0.0.1	Serial0/0
1.1.1.2	0	FULL/ -	00:00:36	30.0.0.1	Serial0/1

```
Router#
```

Task 5.

**Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table.

Hint: Use show ip route ospf to see route changes after modifying the cost.**

Router 1

```
Router#show ip route ospf  
O    30.0.0.0 [110/128] via 10.0.0.2, 00:13:52, Serial0/1  
      [110/128] via 20.0.0.2, 00:13:52, Serial0/0  
  
Router#
```

Router 2

```
Router#show ip route ospf
O    10.0.0.0 [110/128] via 20.0.0.1, 00:16:27, Serial0/0
      [110/128] via 30.0.0.2, 00:16:27, Serial0/1
Router#
```

Router 3

```
Router#show ip route ospf
O    20.0.0.0 [110/128] via 10.0.0.1, 00:18:00, Serial0/0
      [110/128] via 30.0.0.1, 00:18:00, Serial0/1
Router#
```
